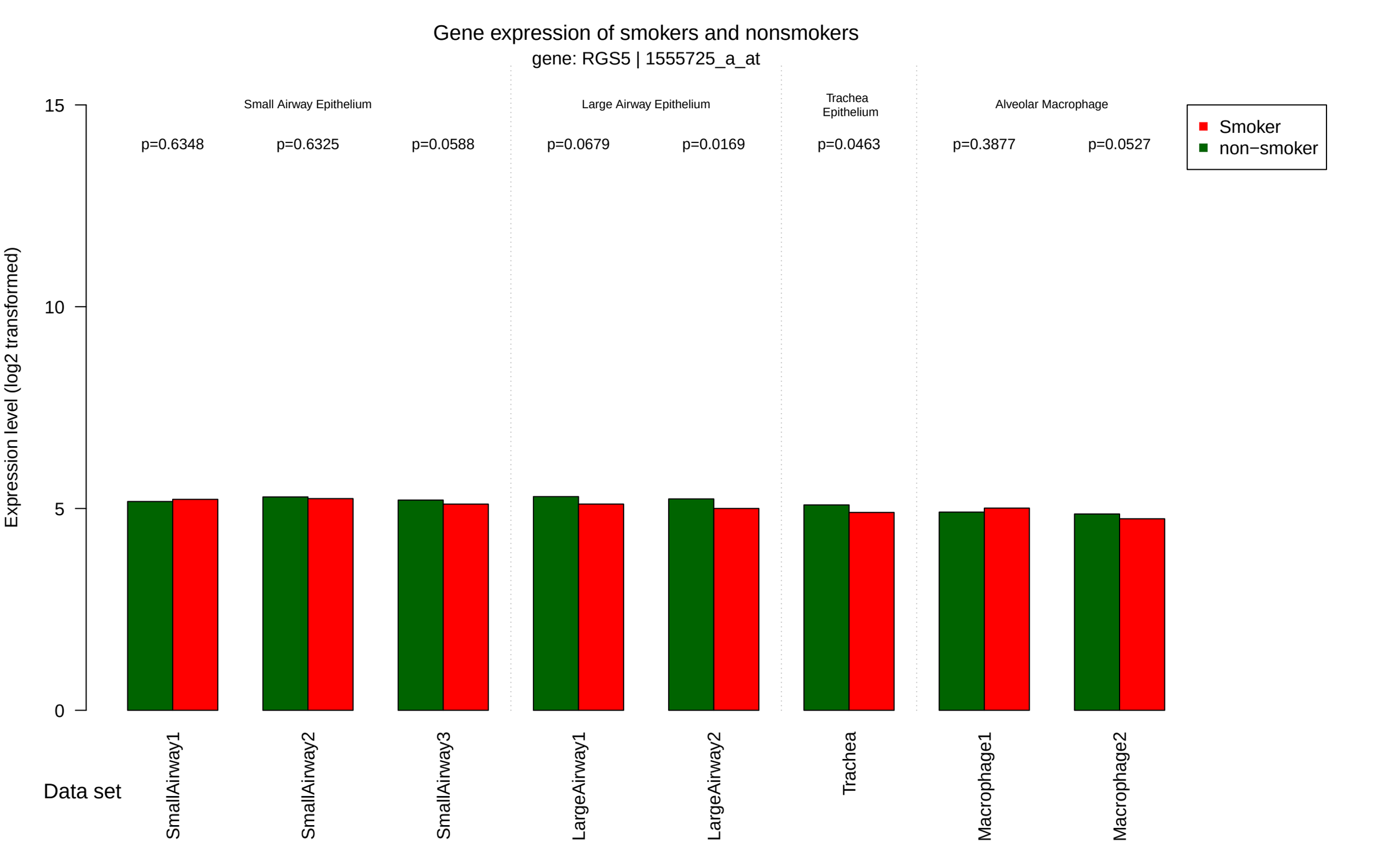
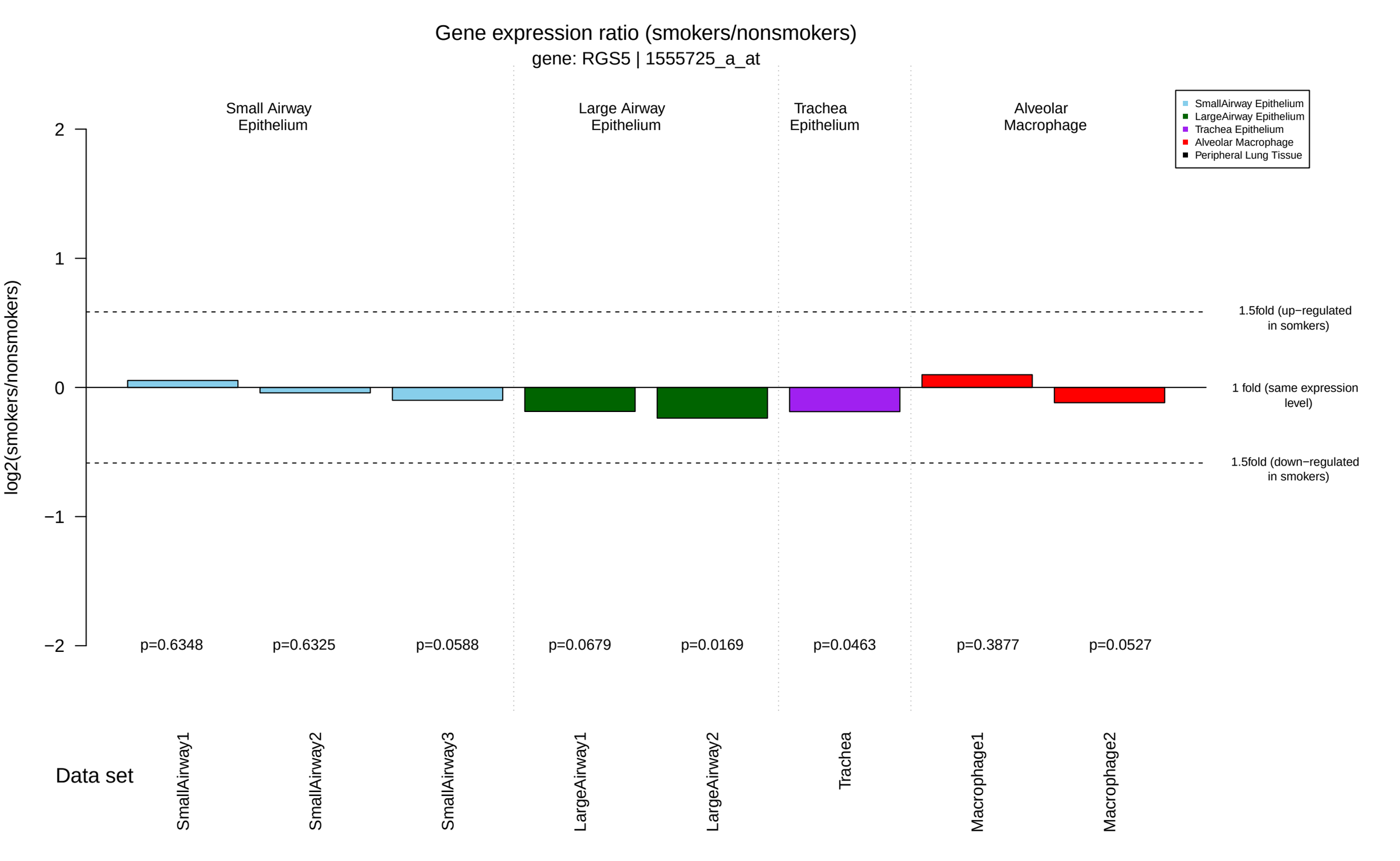
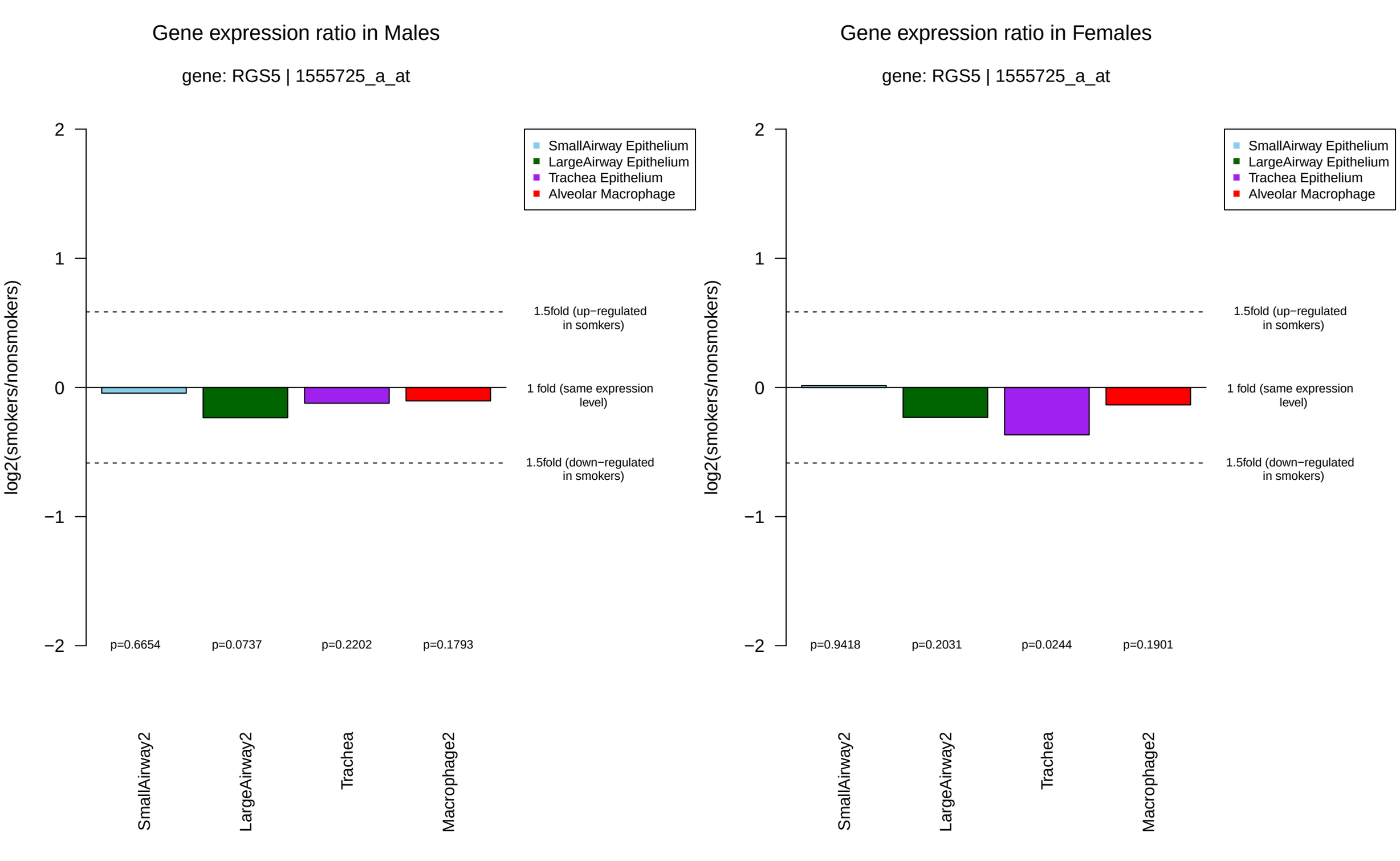
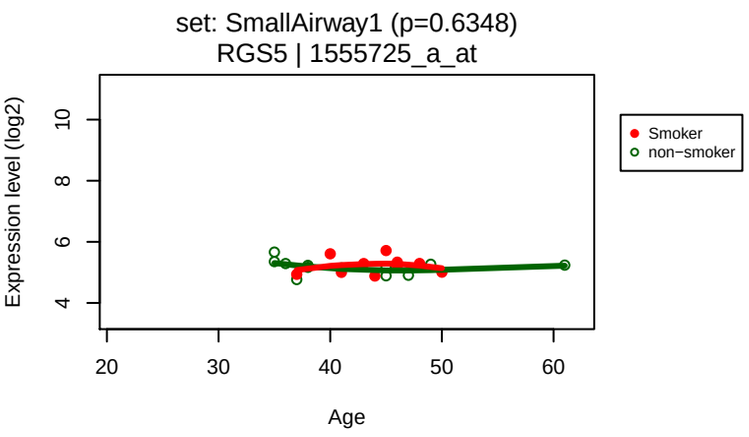
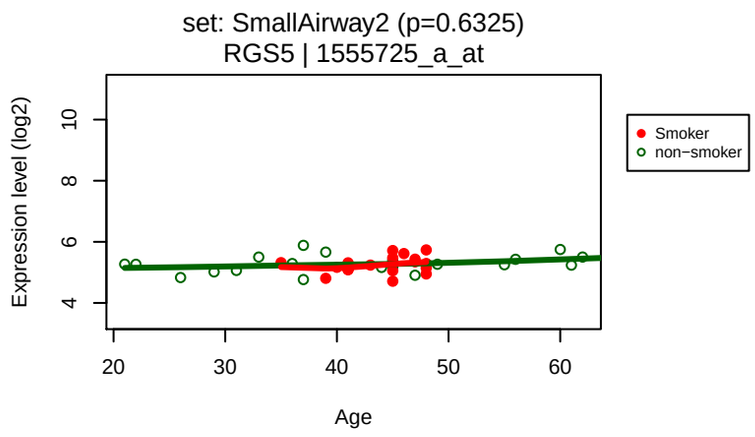
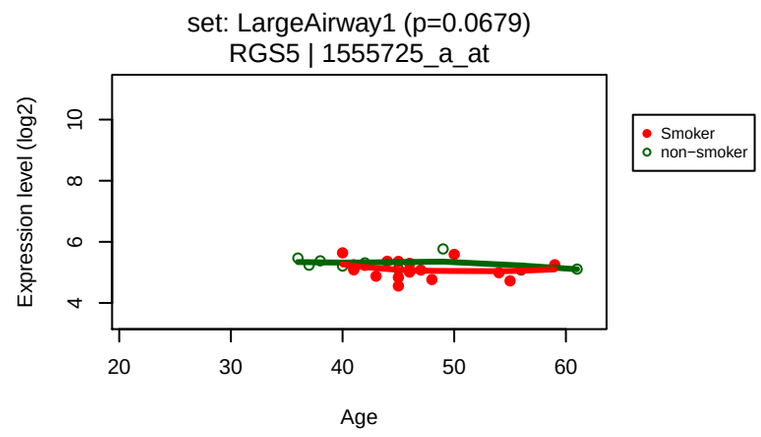

Gene expression of smokers and nonsmokers
gene: RGS5 | 1555725_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6348
p=0.6325
p=0.0588
p=0.0679
p=0.0169
p=0.0463
p=0.3877
p=0.0527
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

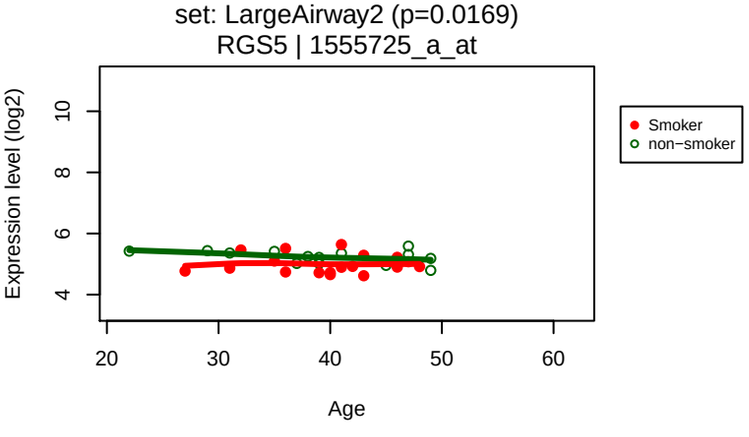

Gene expression ratio (smokers/nonsmokers)
gene: RGS5 | 1555725_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6348
p=0.6325
p=0.0588
p=0.0679
p=0.0169
p=0.0463
p=0.3877
p=0.0527
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

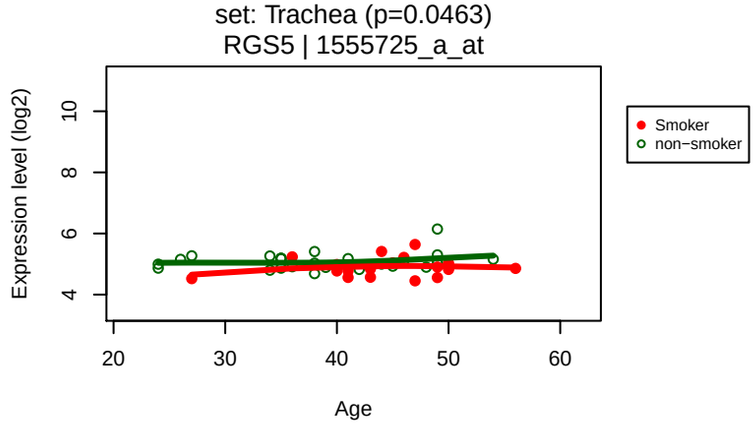

Gene expression ratio in Males
gene: RGS5 | 1555725_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6654
p=0.0737
p=0.2202
p=0.1793
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS5 | 1555725_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9418
p=0.2031
p=0.0244
p=0.1901
SmallAirway2
LargeAirway2
Trachea
Macrophage2

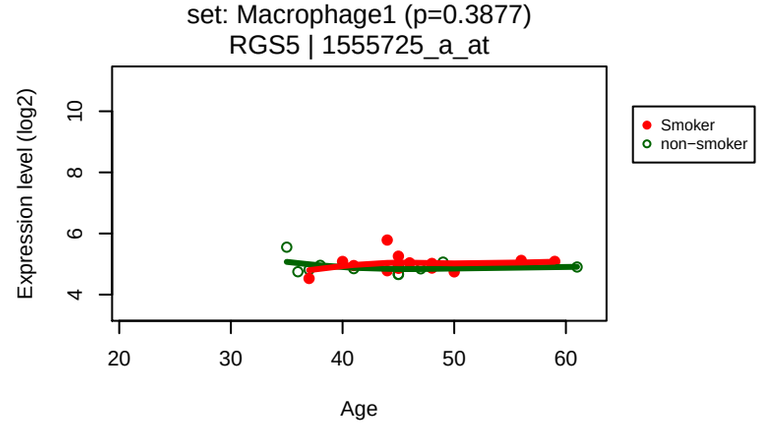

set: SmallAirway1 (p=0.6348)
RGS5 | 1555725_a_at
Expression level (log2)
Age
Smoker
non−smoker

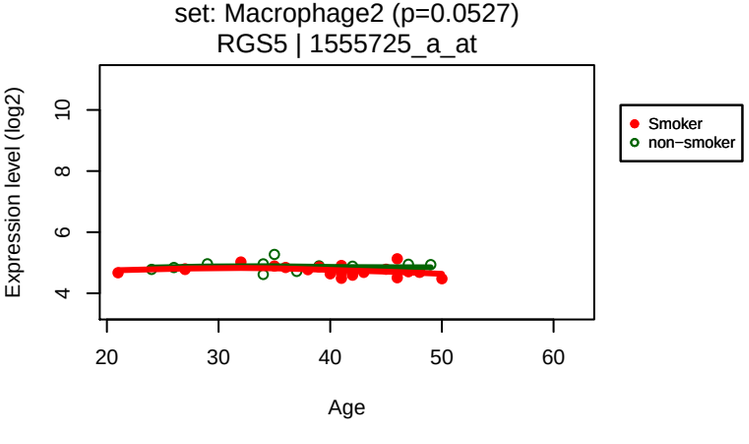

set: SmallAirway2 (p=0.6325)
RGS5 | 1555725_a_at
Expression level (log2)
Age
Smoker
non−smoker

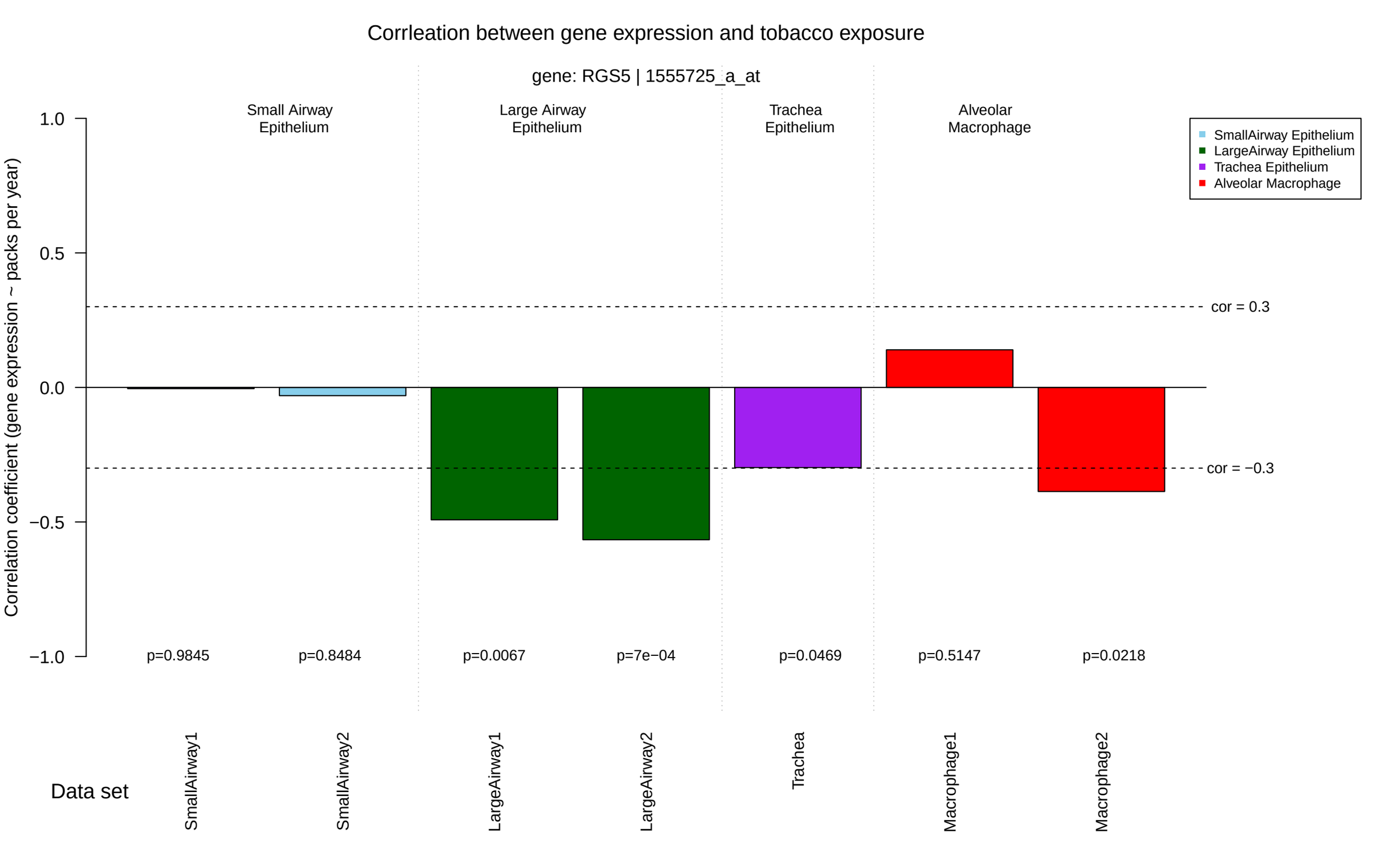

set: LargeAirway1 (p=0.0679)
RGS5 | 1555725_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0169)
RGS5 | 1555725_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0463)
RGS5 | 1555725_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3877)
RGS5 | 1555725_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0527)
RGS5 | 1555725_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS5 | 1555725_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9845
p=0.8484
p=0.0067
p=7e−04
p=0.0469
p=0.5147
p=0.0218
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2